

Coffee Kiosk POS - Customer Frontend

Test Plan

Document field	Value
Project	Coffee Kiosk POS
Module	Customer - Frontend
Reference documents	FRS v1.0 and Test Estimation Report v1.0
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This plan becomes the active QA baseline after the referenced FRS and this Test Plan are approved. Material requirement changes require impact analysis and plan updates.

1. Purpose

This Test Plan defines the scope, objectives, approach, resources, schedule, controls, and completion criteria for manual testing of the Coffee Kiosk POS customer-facing frontend.

2. Test Objectives

- Verify all 48 requirements defined in the customer-frontend FRS.
- Confirm that customers can complete the flow from menu browsing to receipt generation.
- Validate order items, quantities, taxes, coupon discounts, and payable totals.
- Verify customer-visible UPI and Card payment outcomes and recovery paths.
- Confirm session timeout, privacy, usability, reliability, and role-access requirements.
- Identify and report defects before release approval.

3. Scope

3.1 Features to test

Feature	Main coverage
Food Catalog	Categories, item details, availability, item options, adding items, and cart updates
Cart	Item display, quantity changes, removal, totals, empty cart, and navigation
Checkout	Locked order, dine-in/takeaway, coupon validation, totals, and payment-method selection
Payment Gateway	UPI/Card navigation, amount and order reference, processing, success, failure, timeout, cancellation, and retry
Confirmation and Receipt	Order details, bill calculation, emailed receipt, and starting a new session
Non-functional behaviour	Touch usability, accessibility basics, privacy, session security, reliability, and kiosk compatibility

3.2 Features not to test

- Brew Crew and IT Admin interfaces.
- Direct API, database, AWS, and backend component testing.
- Internal functionality of the third-party payment provider.
- Load, stress, and full performance testing.
- Penetration testing or security certification.
- Test automation.
- KDS, thermal printer, cash payment, and physical card-terminal functionality.
- Devices, resolutions, and browsers not approved for the kiosk release.

4. Test Approach and Strategy

4.1 Overall approach

Testing will use a manual, black-box, requirement-based approach. Each FRS requirement will be mapped to test scenarios and test cases through the Requirement Traceability Matrix (RTM).

Test design will include:

- Positive and negative scenarios.
- Boundary and validation checks.
- Navigation and end-to-end customer flows.
- Integration failure and recovery paths.
- Error messages and prevention of duplicate actions.

4.2 Test types

Test type	Purpose
Smoke testing	Confirm that the build and main order flow are ready for detailed testing.
Functional testing	Validate behaviour against the FRS.
Integration testing	Validate customer-visible interaction with backend, payment, coupon, and email services.
UI and usability testing	Check touch controls, labels, feedback, readability, and navigation consistency.
Basic security and privacy testing	Validate role protection, session clearing, and handling of payment and email data.
Compatibility testing	Validate the approved landscape kiosk resolution and supported Chromium-based browser.
Retesting	Verify that reported defects have been fixed.
Regression testing	Confirm that fixes have not affected the main customer order flow.

4.3 Manual versus automation

All planned test design and execution will be manual. Automation assessment and implementation are outside this release scope.

4.4 Test evidence and traceability

- Each test case will record expected result, actual result, status, and supporting evidence where needed.
- Allowed execution statuses are **Pass**, **Fail**, **Blocked**, and **Not Executed**.
- RTM links will follow: **FRS Requirement** → **Test Scenario** → **Test Case** → **Result** → **Defect**.

5. Test Environment

Area	Requirement
Customer device	Touchscreen kiosk or representative landscape test device
Browser	Supported modern Chromium-based kiosk browser
Frontend	ReactJS QA build
Supporting services	Spring Boot test backend and AWS-managed test data services
Payment	Approved UPI and Card payment sandbox with test credentials
Test data	Available/unavailable items and valid, invalid, expired, and inapplicable coupons
Email	Controlled test inbox for receipt delivery
Network	Stable connection to required QA services

6. Roles and Responsibilities

Role	Responsibility
Project Manager	Approves schedule, coordinates release decisions, and resolves project-level blockers.
QA Lead	Reviews the Test Plan, scenarios, cases, risks, defects, and closure recommendation.
QA Engineer	Prepares test data and documents; executes tests; logs defects; performs retesting and regression; maintains RTM and reports.
Developer / Integration Team	Provides testable builds, investigates defects, supplies fixes, and supports integrations.
Product / Business Stakeholder	Clarifies requirements, confirms expected business behaviour, and accepts residual risk.

7. Test Schedule

The schedule follows the approved Test Estimation Report and assumes one full-time QA Engineer.

Phase	Planned duration
Requirement analysis and test planning	1-2 working days
Test scenario and test-case design	3-4 working days
Review, environment validation, and smoke testing	1 working day
Test execution and defect reporting	4-5 working days
Retesting, regression, RTM, and closure	2-3 working days
Total planned duration	Approximately 14 working days

The schedule may change if builds, fixes, approvals, or third-party services are delayed.

8. Entry and Exit Criteria

8.1 Entry criteria

- FRS, Test Plan, scenarios, and test cases are reviewed and approved.
- The QA build is deployed and release notes or known limitations are available.
- Required environment, menu data, coupon data, payment sandbox, and email inbox are ready.
- Unit and integration checks owned by the development team are completed.
- Smoke testing confirms that the main customer order flow is testable.

8.2 Exit criteria

- All planned test cases are executed or formally dispositioned.
- All FRS requirements have traceable test coverage.
- All P0 and P1 test cases pass.
- No open Critical or High defect remains unless formally accepted by stakeholders.
- Planned defect retesting and regression testing are complete.
- RTM, execution report, defect report, and test closure report are updated.
- Residual risks are documented and accepted before release recommendation.

9. Suspension and Resumption Criteria

9.1 Suspension criteria

Testing may be suspended when:

- Smoke testing fails on a release-blocking customer flow.
- The QA environment, backend, payment sandbox, or essential test data is unavailable.
- Order or payment results are unreliable and cannot be verified.
- A Critical defect prevents meaningful continuation of planned testing.
- The build differs materially from the approved requirements.

9.2 Resumption criteria

Testing may resume when:

- A corrected build or restored service is available.
- The blocking issue has been validated through smoke testing.
- Required test data and dependencies are usable.
- The QA Lead confirms the revised execution sequence and schedule impact.

10. Defect Management Process

1. Reproduce the issue and collect clear evidence.
2. Log the defect in Jira with summary, environment, steps, expected result, actual result, severity, priority, and attachments.
3. Review defects during triage with QA, development, and project stakeholders.
4. Assign the defect for correction or an approved disposition.
5. Retest the supplied fix and update the result.
6. Perform regression testing where the change may affect related flows.

7. Close, reopen, defer, or reject the defect with documented justification.

Defect workflow:

New → Assigned → In Progress → Fixed → Retest → Closed / Reopened / Deferred

Severity	Meaning
Critical	Payment, order completion, or the application is unusable with no workaround.
High	A major customer feature fails or produces incorrect order/payment behaviour.
Medium	Functionality is affected, but a reasonable workaround exists.
Low	Minor UI, wording, or low-impact usability issue.

11. Tools

Purpose	Tool or format
Test documentation	Markdown and spreadsheet files
Defect tracking	Jira
Browser inspection	Chromium DevTools
Payment testing	Approved gateway sandbox
Version control	GitHub
Communication and approval	Email / Jira

12. Risks and Mitigation

Risk	Impact	Mitigation
Payment sandbox is unstable	Payment cases become blocked	Confirm access before execution and coordinate approved simulated outcomes if available.
Menu, coupon, or email data is unavailable	Related scenarios cannot be completed	Prepare and verify test data before the test cycle.
QA build is unstable	Execution is delayed and results become unreliable	Apply entry criteria and return failed builds for correction.
FRS changes after approval	Existing coverage and schedule become incomplete	Perform change impact analysis and update estimates, cases, and RTM.
High defect density or slow fixes	Retesting exceeds the planned schedule	Prioritise P0/P1 flows and communicate when contingency is consumed.

13. Test Deliverables

Before execution

- Approved FRS.
- Test Estimation Report.
- Test Plan.
- Test Scenarios and Test Cases.
- Test data and environment-readiness confirmation.

During execution

- Test execution results and evidence.
- Defect reports.
- Updated RTM.
- Test progress updates.

After execution

- Final Test Execution Report.
- Final Defect Report.
- Completed RTM.
- Test Summary and Closure Report.
- Release recommendation with residual risks.

14. Approvals and Sign-off

Stakeholder	Representative	Approval method	Date	Status
Prepared by	Kushagra Sinha (QA Engineer)	Email	25 September 2026	Completed
QA Lead	To be assigned	Email / Jira	-	Pending
Project Manager	To be assigned	Email / Jira	-	Pending
Product / Business Stakeholder	To be assigned	Email / Jira	-	Pending